

Formulas

Formulas and Equations

2026

Preliminary Mathematics Standard

Outline

Substitution into Algebraic Expressions

Evaluating the Subject of a Formula

Distance, Speed and Time

Blood Alcohol Content (BAC)

Medication Dosage Formulas

Changing the Subject of a Formula

Homework

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Substitution into Linear Expressions

Key Skill: Replace pronumerals with given values and evaluate.

Example 1: Evaluate $3x + 5$ when $x = 4$

$$\begin{aligned}3x + 5 &= 3(4) + 5 \\&= 12 + 5 \\&= 17\end{aligned}$$

Example 2: Evaluate $2a - 3b$ when $a = 7$ and $b = 2$

$$\begin{aligned}2a - 3b &= 2(7) - 3(2) \\&= 14 - 6 \\&= 8\end{aligned}$$

Substitution into Non-Linear Expressions

Example 1: Evaluate $x^2 + 3x - 2$ when $x = -3$

$$\begin{aligned}x^2 + 3x - 2 &= (-3)^2 + 3(-3) - 2 \\&= 9 - 9 - 2 \\&= -2\end{aligned}$$

Example 2: Evaluate $\frac{a^2 - b^2}{a + b}$ when $a = 5$ and $b = 3$

$$\begin{aligned}\frac{a^2 - b^2}{a + b} &= \frac{(5)^2 - (3)^2}{5 + 3} \\&= \frac{25 - 9}{8} \\&= \frac{16}{8} = 2\end{aligned}$$

Practice: Substitution

Evaluate the following expressions:

Example

1. $5x - 7$ when $x = 3$
2. $x^2 - 4x + 1$ when $x = 2$
3. $\frac{2p+q}{3}$ when $p = 6$ and $q = 3$
4. $a^2 + 2ab + b^2$ when $a = 4$ and $b = -1$

Practice: Substitution

Evaluate the following expressions:

Example

1. $5x - 7$ when $x = 3$
2. $x^2 - 4x + 1$ when $x = 2$
3. $\frac{2p+q}{3}$ when $p = 6$ and $q = 3$
4. $a^2 + 2ab + b^2$ when $a = 4$ and $b = -1$

Answers: 1) 8 2) -3 3) 5 4) 9

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Finding the Subject of a Formula

The subject is the pronumeral we are solving for (usually on the left side).

Example 1: Area of a rectangle: $A = lw$

- If $l = 8$ cm and $w = 5$ cm, find A
- $A = 8 \times 5 = 40 \text{ cm}^2$

Example 2: Perimeter of a rectangle: $P = 2(l + w)$

- If $l = 12$ m and $w = 7$ m, find P
- $P = 2(12 + 7) = 2(19) = 38 \text{ m}$

More Complex Formulas

Volume of a cylinder: $V = \pi r^2 h$

Example

Find V when $r = 3$ cm and $h = 10$ cm

$$\begin{aligned} V &= \pi(3)^2(10) \\ &= \pi \times 9 \times 10 \\ &= 90\pi \\ &\approx 282.74 \text{ cm}^3 \end{aligned}$$

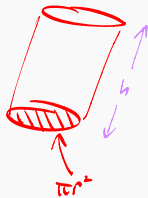


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Distance, Speed and Time Formulas

Three interconnected formulas:

$$D = S \times T$$

$$S = \frac{D}{T}$$

$$\frac{\text{(m)}}{\text{(s)}} \rightarrow \text{m/s}$$

$$T = \frac{D}{S}$$

where:

- D = distance (m)
- S = speed (m/s, ms⁻¹)
- T = time (s)

Remember: Units must be consistent!

Example: Distance Calculation

Example

A car travels at 80 km/h for 2.5 hours. How far does it travel?

Solution:

$$\begin{aligned} D &= S \times T \\ &= 80 \times 2.5 \\ &= 200 \text{ km} \end{aligned}$$

Example: Unit Conversion

Example

A cyclist travels 45 km in 2 hours and 30 minutes. What is their average speed?

Solution:

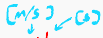
- First convert time: 2 hours 30 min = 2.5 hours
- Then use formula:

$$\begin{aligned} S &= \frac{D}{T} \\ &= \frac{45}{2.5} \\ &= 18 \text{ km/h} \end{aligned}$$

Stopping Distance Formula

Total Stopping Distance = Reaction Distance + Braking Distance

Common formulas:

- **Reaction distance:** $d_r = \frac{vt}{3.6}$ or $d_r = vt$ 
- where v is speed in km/h, t is reaction time in seconds
- **Braking distance:** $d_b = \frac{v^2}{2gf}$ or simplified: $d_b = \frac{v^2}{170}$ (dry road)
- where v is speed in km/h

Note: Stopping distance increases dramatically with speed!

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BAC Formula

Blood Alcohol Content (BAC) can be estimated using:

For males:

$$\text{BAC} = \frac{10N - 7.5H}{6.8M}$$

For females:

$$\text{BAC} = \frac{10N - 7.5H}{5.5M}$$

where:

- N = number of standard drinks consumed
- H = number of hours of drinking
- M = person's weight in kilograms

$$\text{BAC} \propto N$$

As $N \uparrow$, BAC $\uparrow$

$$\text{BAC} \propto \frac{1}{H}$$

As $H \uparrow$, BAC $\downarrow$

$$\text{BAC} \propto \frac{1}{M}$$

BAC Example

Example

A 75 kg male consumes 5 standard drinks over 3 hours. What is his estimated BAC?

Solution:

$$\begin{aligned}\text{BAC} &= \frac{10N - 7.5H}{6.8M} \\ &= \frac{10(5) - 7.5(3)}{6.8(75)} \\ &= \frac{50 - 22.5}{510} \\ &= \frac{27.5}{510} \\ &\approx 0.054\end{aligned}$$

This is **over** the legal limit of 0.05!

$$T = \frac{BAC}{0.015}$$

Time to Zero BAC

Formula for hours to reach zero BAC:

$$H = \frac{10N - 6.8M \times \text{BAC}}{7.5}$$

(for males; similar formula for females with 5.5 instead of 6.8)

Example

How many hours must the person from the previous example stop drinking to reach zero BAC?

$$\begin{aligned} H &= \frac{10(5) - 6.8(75)(0)}{7.5} \\ &= \frac{50}{7.5} \\ &\approx 6.67 \text{ hours} \end{aligned}$$

Limitations of BAC Estimation

Important limitations:

1. **Individual variation:** Metabolism rates differ between people
2. **Food consumption:** Food slows alcohol absorption
3. **Type of alcohol:** Different drinks are absorbed at different rates
4. **Gender differences:** Women typically metabolize alcohol differently
5. **Body composition:** Muscle vs fat ratio affects BAC
6. **Medications:** Can interact with alcohol metabolism
7. **Health conditions:** Liver function affects processing
8. **Time assumptions:** Formulas assume uniform drinking rate

Conclusion: These are *estimates only*. Never rely on calculations to determine if safe to drive!

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Pediatric Dosage Formulas

Three main formulas for calculating children's medication doses:

1. Fried's Formula (for infants under 2 years):

$$\text{Child dose} = \frac{\text{Age (months)} \times \text{Adult dose}}{150}$$

2. Young's Formula (for children 1-12 years):

$$\text{Child dose} = \frac{\text{Age (years)} \times \text{Adult dose}}{\text{Age (years)} + 12}$$

3. Clark's Formula (based on weight):

$$\text{Child dose} = \frac{\text{Weight (kg)} \times \text{Adult dose}}{70}$$

Assume ave. adult weight is 70 kg.

$$\text{Child dose} = \frac{\text{weight (kg)}}{70 \text{ (kg)}} \times \text{Adult dose}$$

Medication Dosage Example 1

Example

The adult dose of a medication is 500 mg. What dose should be given to a 6-year-old child using Young's formula?

Solution:

$$\begin{aligned}\text{Child dose} &= \frac{\text{Age} \times \text{Adult dose [mg]}}{\text{Age} + 12} \\ &= \frac{6 \times 500}{6 + 12} \\ &= \frac{3000}{18} \\ &\approx 167 \text{ mg}\end{aligned}$$

Medication Dosage Example 2

Example

An 8-month-old infant needs medication with an adult dose of 400 mg. Calculate the dose using Fried's formula.

Solution:

$$\begin{aligned}\text{Child dose} &= \frac{\text{Age (months)} \times \text{Adult dose}}{150} \\ &= \frac{8 \times 400}{150} \\ &= \frac{3200}{150} \\ &\approx 21.3 \text{ mg}\end{aligned}$$

Medication Dosage Example 3

Example

A child weighing 25 kg requires medication with an adult dose of 600 mg. Use Clark's formula.

Solution:

$$\begin{aligned}\text{Child dose} &= \frac{\text{Weight (kg)} \times \text{Adult dose}}{70} \\ &= \frac{25 \times 600}{70} \\ &= \frac{15000}{70} \\ &\approx 214 \text{ mg}\end{aligned}$$

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Changing the Subject: Linear Formulas

Goal: Rearrange formula to make a different pronumeral the subject.

Example 1: Make w the subject of $P = 2(l + w)$

$$P = 2(l + w)$$

$$\frac{P}{2} = l + w \quad \text{(divide both sides by 2)}$$

$$\frac{P}{2} - l = w \quad \text{(subtract } l \text{ from both sides)}$$

$$w = \frac{P}{2} - l \quad \text{(rearrange)}$$


subject
on LHS

Changing the Subject: More Examples

Example 2: Make h the subject of $V = \pi r^2 h$

$$V = \pi r^2 h$$

$$\frac{V}{\pi r^2} = h \quad (\text{divide both sides by } \pi r^2)$$

$$h = \frac{V}{\pi r^2}$$

Example 3: Make t the subject of $v = u + at$

$$v = u + at$$

$$v - u = at \quad (\text{subtract } u)$$

$$\frac{v - u}{a} = t \quad (\text{divide by } a)$$

$$t = \frac{v - u}{a}$$

Changing the Subject: Quadratic Forms

For quadratics of the form $y = ax^2 + bx + c$

Example

Make ~~r~~ the subject of $A = \pi r^2$
 r

$$A = \pi r^2$$

$$\frac{A}{\pi} = r^2$$

(divide by π)

$$\sqrt{\frac{A}{\pi}} = r$$

(take square root)

$$r = \sqrt{\frac{A}{\pi}}$$

($r > 0$, length)

Note: We take the positive square root for physical measurements.

Practice: Changing the Subject

Make the indicated pronumeral the subject:

Example

1. $A = \frac{1}{2}bh$ (make h the subject)
2. $C = 2\pi r$ (make r the subject)
3. $E = \frac{1}{2}mv^2$ (make v the subject)
4. $S = ut + \frac{1}{2}at^2$ (make a the subject)

Practice: Changing the Subject

Make the indicated pronumeral the subject:

Example

1. $A = \frac{1}{2}bh$ (make h the subject)
2. $C = 2\pi r$ (make r the subject)
3. $E = \frac{1}{2}mv^2$ (make v the subject)
4. $S = ut + \frac{1}{2}at^2$ (make a the subject)

Answers:

1. $h = \frac{2A}{b}$
2. $r = \frac{C}{2\pi}$
3. $v = \sqrt{\frac{2E}{m}}$
4. $a = \frac{2(S-ut)}{t^2}$

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Complete the following questions from Fitzpatrick New Senior Mathematics Exercise 4.8 Pg. 110

- Question 1
- Question 2
- Question 3